Vishal Sharma

(+49) 1523 1306707 | writetovishal.sharma@gmail.com
Stuttgart, Germany | Blue-Card holder | linkedin.com/in/vishal-sharma

ABOUT ME

Battery Domain Expert | Simulation & Modeling Expertise | 15 years of working experience

Work experience spanning across Tier 1 cell manufacturer and premium automotive OEM R&Ds - A123 Systems, Mercedes-Benz, Volvo Trucks and Caterpillar. Specializing in Li-ion cell technology, systems modeling, battery pack development, development process automation and cross-functional technical leadership. Developed in-house production-ready battery and full-vehicle simulation frameworks/toolchains deployed in Mercedes, Volvo and A123. Devised technical roadmaps and RFQ winning strategies across cross-timezone DE/US/CN divisions.

Simulation Teams Built: 3 | Engineers Led: 10+ | Projects Scaled: 7x | Model Dev. Time Reduced: ~60%

CORE COMPETENCIES

Battery Systems: Pack development, Cell-to-Pack scaling, Thermal & Ageing modelling(calendaric and cyclic ageing), battery management systems, LV & HV automotive battery, ESS/BESS battery systems, ISO 26262(functional safety).

Technical methods - digital twin, electrochemical & EEC modeling/simulation, cell characterisation & parametrisation, P2D/SPM/DFN modeling, BMS algorithms, SoC/SoH estimation, EKF/UKF state estimation methods, PyBaMM.

Simulation & Modeling: Component & system level modeling, physics-based modeling, systems engineering(MBSE), simulation framework/toolchain development, SiL/MiL framework, MATLAB/Simulink, Python, ML/AI methods - PINN.

FEM: ANSYS, ABAQUS, HYPERMESH, FEMFAT, ParaView - thermal, structural, multiphysics

Leadership & Delivery: 8 years of leadership experience, Simulation team building (x3), Technical roadmap definition, Cross-timezone EMEA coordination(DE/US/CN), Requirements Engineering, RFI/RFQ technical strategy, IBM DOORS.

WORK EXPERIENCE

A123 Systems GmbH, Germany

Simulation Lead, EMEA | Nov 2022 - Present

- **Technical Governance:** Defined and cascaded simulation deliverables from OEM RFI/RFQ specifications - with full responsibility and final validation authority for all EMEA technical deliverables(6 major RFQs, 15+ RFI).
- **Reduced RFI/RFQ response time by ~30%** - by defining simulation requirements clearly, aligning A123 China HQ & US teams across time zones and functions, identifying/resolving the bottlenecks and eliminating iteration loops.
- **Won 2 major OEM 48V projects**(high volume/multi-million euro contract) - by thorough assessments of technical deliverables, thorough simulation-driven feasibility studies, extensive electrical, thermal and aging test data analysis.
- **Cell Portfolio Enhancement:** Provided simulation-driven feasibility analysis for next-gen **cell concept development** - optimising chemistry and form factor variants under performance, manufacturing and cost constraints.
- **Engineering solutions:** self-initiated tools to address technical problems- **Current limits generator**(Li-plating, aging-aware constraints), **Virtual cell scaling**, pseudo-3D electro-thermal pack model and Pack Cost estimator.
- Developed Data Compilation Manager tool for large-scale test-data compilation for multiple projects & cell variants
- Developed end-to-end pack concept for multiple RFI/RFQ specifications (12 V/24 V/48 V battery packs) - covering cell selection, packaging, BMS/EE integration, thermal management, and functional safety (ISO 26262) compliance.

Volvo Trucks R&D, India

Lead – Simulation & Analysis (Senior Manager) | Mar 2022 - Sep 2022

- **Technical lead** for **10-person multidisciplinary team** for thermal, charging, SIL/HIL, and full-vehicle simulation & modelling workstreams (including VECTO compliance) - sharing responsibilities with Swedish counterparts.
- Developed rapid-execution **PINN-based battery degradation** and **charging model** for Volvo fleet mobile app- for quick evaluation of truck fleets and route optimization; model accuracy ~90%, computation time in seconds.
- Developed standalone battery simulation framework for focused battery & BMS simulations-run time significantly.

Caterpillar R&D, India

Performance Engineer | Sep 2021 - Feb 2022

- **Set up a battery modeling & simulation team** in the new EV department in India R&D division.
- Set up **Cell Testing Facility** end-to-end - from cell procurement to setting-up testing procedures and identifying testing equipment capabilities, coordinating with logistics, legal and purchase of cross-site divisions (IN/US).
- Developed empirical degradation models for NMC/LFP chemistries from experimental data; conducted parameter estimation and identification for battery model development, and extensive thermal analysis on test data.

Mercedes-Benz R&D India Ltd.

Technology Lead (Domain Expert) | Jun 2018 - Sep 2021

- **Battery and Lithium-ion cell SME** across cross-functional collaborative projects; Individual Contributor on **high complexity and specialized in-house battery modeling** processes/tools alongside team leadership.

- **5x Business Growth:** Built Daimler counterpart collaboration through regular German visits, proactively capturing business requirements; *portfolio grew from 1 to 7 parallel projects, team size expanded to 4 through direct hiring*.
- Developed **Battery Thermal Model Configurator** - reduced model development time by ~**60%** - generating high-fidelity parameterised coupled CFD-thermal ROM for HV battery systems(faster & accurate thermal model).
- Led the development/enhancement of **battery modelling toolchain** end-to-end - from defining strategy, architectural design of tools/processes, technical review and project planning till release of stable versions *(Audit-ready)*.

Senior Engineer (Top Expert) | Jun 2015 - Jun 2018

- Built EV simulation capability on top of existing ICE in-house **full-vehicle simulation framework** - modelled subsystems and component models for different vehicle configurations (E-Class, S-Class, AMG, & EQC).
- Developed a high accuracy **thermal model of DC-DC converter** to solve the critical problem of overheating in field vehicles and validated the modelled components against experiments with **96% accuracy** against test data.
- Developed and validated **BMS algorithms** (SOC estimation, current limitations, thermal protection | MiL framework) & integrated into vehicle-level simulation for virtual validation; built **SiL framework** as a parallel branch.
- Expanded simulation framework to EV/HEV/PHEV architecture, by developing & integrating **ADAS** functionality, **BMS logic**, **Charging** and **Thermal Management System** modules, enabling virtual development from a unified platform.
- Conducted full-vehicle and component-level performance simulations for different EV/HEV/PHEV configurations and different subsystems configurations at pre-concept, concept, and post-production stages for architectural decisions.

Hero MotoCorp Ltd., India

Deputy Manager, R&D - Engine Design Group | Nov 2013 - May 2015

- Performed 1D/3D engine modelling- non-linear structural, thermal-structural, and fatigue analyses.
- Optimized front fender design for aerodynamic drag by surrogate modeling in a cross-functional project.
- Developed a new analysis methodology to optimise engine mounts for engine vibration reduction.

TVS Motor Company, India

Member, R&D - Design & Analysis Group | Nov 2011 - Oct 2013

- Conducted Linear and non-linear structural analyses of chassis components & topology optimisation.
- Developed ANSYS APDL Macros reducing pre-processing and analysis time by ~50%.

Victoria University, Melbourne, Australia

Internship - Optimised material properties for CFD fire modelling using **Genetic Algorithms** | Jun 2009 - Aug 2009

EDUCATION

Master of Technology (M.Tech) - Mechanical Engineering | IIT Kanpur, India | Jun 2010 - Aug 2011

Thesis: Critical Velocity and Standing Waves in High-Speed Tires

Bachelor of Technology (B.Tech) - Mechanical Engineering | IIT Kanpur, India | Apr 2006 - Jun 2010

Languages: English (Professional), German (A1/A2 - in progress), Hindi (Native)

PATENTS & PUBLICATIONS

Patent: Cooling system for fuel cell vehicle/EV to improve fuel efficiency (Inventor's Award recipient) | 2016

Patent: Efficient power recapturing from vehicle suspension -energy harvesting mechanism for range extension | 2017

Publication: Estimating wood material properties using optimisation techniques for CFD-based fire modelling | 2010

NOTABLE TOOLS & PROJECTS

Current Limits Generator | Python | A123 Systems

- Developed physics-based BMS protection algorithms and operating envelope for virtual BMS validation and power maps calculation with enhanced current limits, to meet aging requirements.
- Physics-based pack current limit engine - constraint layers: Li-plating (Butler-Volmer), coupled Arrhenius side reactions (SEI, electrolyte oxidation), lumped thermal model (core temperature bottleneck), dI/dt and dT/dt rate limits.

Pseudo-3D electro-thermal pack model | Python | A123 Systems

- Electro-thermal model for battery pack for quick evaluation of concepts of cooling, performance, thermal behaviour, to reduce the concept evaluation time by ~70% and faster pack development.
- Coupled RC model with lumped thermal model with cooling plate- to predict pack performance, thermal behaviour.

Battery Thermal Model Configurator | Matlab | Mercedes-Benz R&D

- Standardized generation of high fidelity coupled CFD-thermal reduced order model (ROM) for HV battery packs- combining cooling plate results from CFD analysis and battery thermal model results from 3D thermal analysis.
- Implemented methods for HTC calculations, power losses, and parameter/input/output mapping to simulink models.
- Adopted across Mercedes HV battery development projects. *Reduced thermal model development time by ~60%*.

REFERENCES

Professional references are available upon request